

Premiga pengtap 673040672-0

Compare the performance for Naive-bayes, Decision-Tree, Linear-SVC,SGD-Classifiers

```
# Import Libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.tree import DecisionTreeClassifier
from sklearn.svm import LinearSVC
from sklearn.linear_model import SGDClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, classification_report
```

```
import pandas as pd
df = pd.read_csv("run_or_walk.csv")
```

```
df = pd.read_csv("run_or_walk.csv")
df.head()
```

	date	time	username	activity	acceleration_x	acceleration_y	acceleration_z	gyro_x	gyro_y	gyro_z
0	2017-6-30	13:51:15:847724020	viktor	0	0.2650	-0.7814	-0.0076	-0.0590	0.0325	-2.9296
1	2017-6-30	13:51:16:246945023	viktor	0	0.6722	-1.1233	-0.2344	-0.1757	0.0208	0.1269
2	2017-6-30	13:51:16:446233987	viktor	0	0.4399	-1.4817	0.0722	-0.9105	0.1063	-2.4367
3	2017-6-30	13:51:16:646117985	viktor	0	0.3031	-0.8125	0.0888	0.1199	-0.4099	-2.9336
4	2017-6-30	13:51:16:846738994	viktor	0	0.4814	-0.9312	0.0359	0.0527	0.4379	2.4922

```
#Define Models
# (For LinearSVC, SGDClassifier, sometimes max_iter=10000 is needed to ensure convergence)
models_to_test = {
    "Gaussian Naive Bayes": GaussianNB(),
    "Decision Tree": DecisionTreeClassifier(random_state=42),
    "Linear SVC": LinearSVC(random_state=42, dual=False, max_iter=10000),
    "SGD Classifier": SGDClassifier(random_state=42, max_iter=10000)
}
```

```
#Prepare Data
# tell the model which part is "features" (X) and which is "target" (y)
X = df.drop(['date', 'time', 'username', 'activity'], axis=1)
y = df['activity']

#Split Data
# test_size=0.2 means 20% of the data goes to testing, 80% to training
# random_state=42 ensures the data split is the same every time we run the code, for consistent results
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

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```
#Train and Store Results
#iterate through each model, train and test it, and store the results.
all_summaries = [] # to store summary results for each model
all_reports = {} # to store full classification reports for each model
```

```
for model_name, model_object in models_to_test.items():
    #Train the model
    model_object.fit(X_train, y_train)

    # predictions
    y_pred = model_object.predict(X_test)

    # Calculate summary
    acc = accuracy_score(y_test, y_pred)
    prec = precision_score(y_test, y_pred, average='weighted', zero_division=0)
    rec = recall_score(y_test, y_pred, average='weighted', zero_division=0)
    f1 = f1_score(y_test, y_pred, average='weighted', zero_division=0)

    #classification report
    report_text = classification_report(y_test, y_pred, zero_division=0)

    # Store the results
    all_summaries.append({
        "Model": model_name,
        "Accuracy": round(acc, 4),
        "Precision": round(prec, 4),
        "Recall": round(rec, 4),
        "F1-Score": round(f1, 4)
    })
    all_reports[model_name] = report_text
```

```
# Display Summary Table
summary_df = pd.DataFrame(all_summaries)
print("\n" + "="*20 + " summary table " + "="*20)
print(summary_df.to_string(index=False)) # prevents displaying row indices
```

```
===== summary table =====
      Model Accuracy Precision Recall F1-Score
Gaussian Naive Bayes   0.9359   0.9425 0.9359   0.9367
      Decision Tree   0.9812   0.9812 0.9812   0.9812
      Linear SVC     0.8657   0.8664 0.8657   0.8660
      SGD Classifier  0.8663   0.8691 0.8663   0.8672
```

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```

===== Classification Reports =====

--- Report for: Gaussian Naïve Bayes ---
      precision    recall  f1-score   support

     0       0.86       0.98       0.92       2785
     1       0.99       0.91       0.95       4771

 accuracy                   0.94       7556
 macro avg       0.93       0.95       0.93       7556
 weighted avg    0.94       0.94       0.94       7556

-----

--- Report for: Decision Tree ---
      precision    recall  f1-score   support

     0       0.98       0.97       0.97       2785
     1       0.98       0.99       0.99       4771

 accuracy                   0.98       7556
 macro avg       0.98       0.98       0.98       7556
 weighted avg    0.98       0.98       0.98       7556

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--- Report for: Linear SVC ---
      precision    recall  f1-score   support

     0       0.81       0.83       0.82       2785
     1       0.90       0.89       0.89       4771

 accuracy                   0.87       7556
 macro avg       0.85       0.86       0.86       7556
 weighted avg    0.87       0.87       0.87       7556

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--- Report for: SGD Classifier ---
      precision    recall  f1-score   support

     0       0.80       0.85       0.82       2785
     1       0.91       0.87       0.89       4771

 accuracy                   0.87       7556
 macro avg       0.85       0.86       0.86       7556
 weighted avg    0.87       0.87       0.87       7556

```

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```
# Find the Best Model
best_model_row = summary_df.loc[summary_df['Accuracy'].idxmax()]
print("\n\n" + "="*20 + " Best Performing Model (by Accuracy)" + "="*20)
print(best_model_row.to_string())
print("="*75)
```

```
===== Best Performing Model (by Accuracy)=====
Model      Decision Tree
Accuracy    0.9812
Precision   0.9812
Recall      0.9812
F1-Score    0.9812
=====
```